

Write a C program to define a structure `Student` and display the details of one student.

Code:

```
#include <stdio.h>

struct Student
{
    char name[20];
    int roll_no;
    float marks;
};

int main()
{
    struct Student s = {"Tejaswini", 101, 89.5};

    printf("Student Details\n");
    printf("Name: %s\n", s.name);
    printf("Roll No: %d\n", s.roll_no);
    printf("Marks: %.2f\n", s.marks);

    return 0;
}
```

Sample output:

Student Details

Name: Tejaswini

Roll No: 101

Marks: 89.50

Write a C program to define a structure of Employee and print the details of three employees.

Code:

```
#include <stdio.h>

struct Employee
{
    char name[20];
    int id;
    float salary;
};

int main()
{
    struct Employee e[3] = {
        {"Ravi", 101, 25000},
        {"Priya", 102, 30000},
        {"Arun", 103, 28000}
    };

    printf("Employee Details\n");
    for(int i = 0; i < 3; i++)
    {
        printf("\nName: %s", e[i].name);
        printf("\nID: %d", e[i].id);
        printf("\nSalary: %.2f\n", e[i].salary);
    }

    return 0;
}
```

Sample output:

Employee Details

Name: Ravi

ID: 101

Salary: 25000.00

Name: Priya

ID: 102

Salary: 30000.00

Name: Arun

ID: 103

Salary: 28000.00

Write a C program to define a structure `Book` and display the book information entered by the user.

Code:

```
#include <stdio.h>
```

```
struct Book
```

```
{
```

```
    char title[50];
```

```
    char author[30];
```

```
    float price;
```

```
};
```

```
int main()
```

```
{
```

```
    struct Book b;
```

```
printf("Enter book title: ");
scanf("%s", b.title);

printf("Enter author name: ");
scanf("%s", b.author);

printf("Enter price: ");
scanf("%f", &b.price);

printf("\nBook Information\n");
printf("Title: %s\n", b.title);
printf("Author: %s\n", b.author);
printf("Price: %.2f\n", b.price);

return 0;
}
```

Sample output:

Enter book title: CProgramming

Enter author name: Dennis

Enter price: 450

Book Information

Title: CProgramming

Author: Dennis

Price: 450.00

Write a C program to define a structure `Product` and calculate the total cost using quantity and unit price.

Code:

```
#include <stdio.h>
```

```
struct Product
```

```
{
```

```
    char name[20];
```

```
    int quantity;
```

```
    float unit_price;
```

```
};
```

```
int main()
```

```
{
```

```
    struct Product p;
```

```
    float total_cost;
```

```
    printf("Enter product name: ");
```

```
    scanf("%s", p.name);
```

```
    printf("Enter quantity: ");
```

```
    scanf("%d", &p.quantity);
```

```
    printf("Enter unit price: ");
```

```
scanf("%f", &p.unit_price);

total_cost = p.quantity * p.unit_price;

printf("\nProduct Details\n");
printf("Name: %s\n", p.name);
printf("Quantity: %d\n", p.quantity);
printf("Unit Price: %.2f\n", p.unit_price);
printf("Total Cost: %.2f\n", total_cost);

return 0;
}
```

Sample output:

Enter product name: Pen

Enter quantity: 10

Enter unit price: 20

Product Details

Name: Pen

Quantity: 10

Unit Price: 20.00

Total Cost: 200.00

Write a C program to define a structure `Rectangle` and calculate its area and perimeter.

Code:

```
#include <stdio.h>
```

```
struct Rectangle
```

```
{
```

```
    float length;
```

```
    float width;
```

```
};
```

```
int main()
```

```
{
```

```
    struct Rectangle r;
```

```
    float area, perimeter;
```

```
    printf("Enter length: ");
```

```
    scanf("%f", &r.length);
```

```
    printf("Enter width: ");
```

```
    scanf("%f", &r.width);
```

```
    area = r.length * r.width;
    perimeter = 2 * (r.length + r.width);

    printf("\nRectangle Details\n");
    printf("Length: %.2f\n", r.length);
    printf("Width: %.2f\n", r.width);
    printf("Area: %.2f\n", area);
    printf("Perimeter: %.2f\n", perimeter);

    return 0;
}
```

Sample output:

Enter length: 10

Enter width: 5

Rectangle Details

Length: 10.00

Width: 5.00

Area: 50.00

Perimeter: 30.00